DO NOW — RECALL

Lesson 089 · The Processor City: Architecture Overview

Name: _____ Class: _____ Date: _____

1. What is the function of a program counter?

2. What does the ALU perform?

3. Name one type of bus.

4. A D flip-flop has D=1 and Q=0 just before the active rising edge. What is Q after the edge?

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

Holds address of next instruction.

2. Model answer

Arithmetic and logical operations.

3. Model answer

Address, data or control bus.

4. Model answer

Q becomes1, as D is sampled at the active edge.

By the end, you can…

• Inside CPU City Hall, the Control Unit organises operations, the ALU carries out arithmetic and logic, and registers hold immediate working state.

• The Memory Warehouse stores program instructions and data. Address, data and control buses connect processor and memory; they carry different kinds of information.

• Von Neumann architecture stores instructions and data in a shared memory/address space, whereas Harvard uses separate instruction and data memories/pathways.

Vocabulary — know these terms

ALU — Arithmetic Logic Unit.

CU — Control Unit.

register — Small fast CPU state storage.

bus — Shared communication pathway.

NEW LEARNING

Read, annotate and explain

Observe

Inside CPU City Hall, the Control Unit organises operations, the ALU carries out arithmetic and logic, and registers hold immediate working state.

Explain

The Memory Warehouse stores program instructions and data. Address, data and control buses connect processor and memory; they carry different kinds of information.

Evaluate

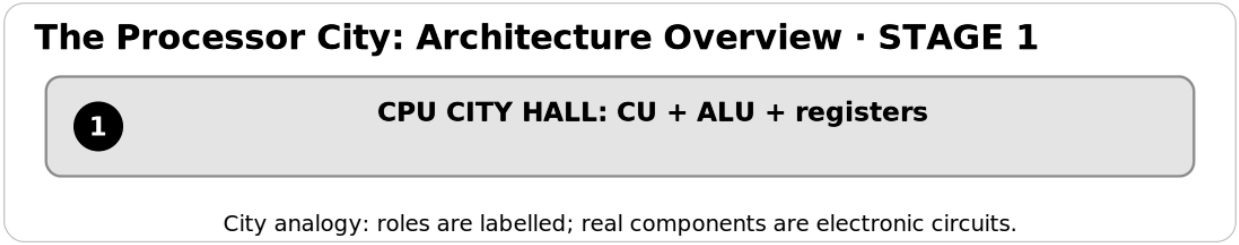
Von Neumann architecture stores instructions and data in a shared memory/address space, whereas Harvard uses separate instruction and data memories/pathways.

Explain one step in your own words, using the correct technical terms.

New Learning · original lesson diagrams

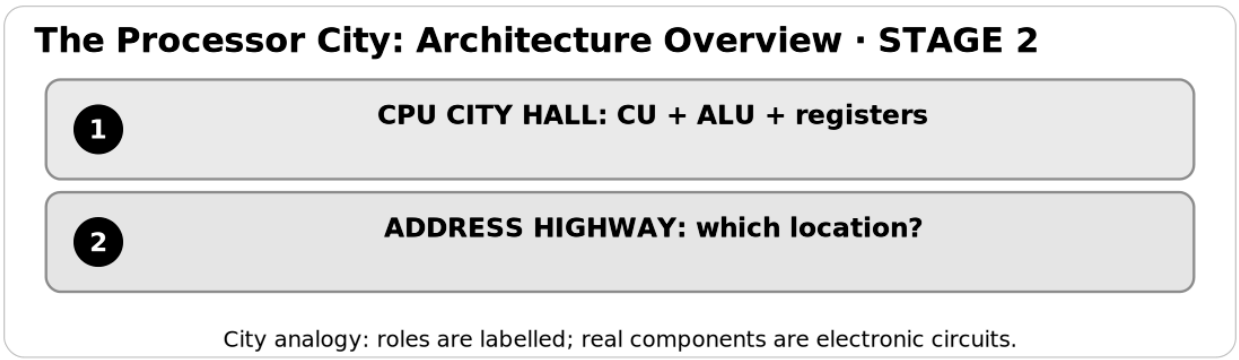
Diagrams appear after the related explanation. Annotate the original source visual.

New Learning



The Processor City: Architecture Overview · STAGE 1; CPU CITY HALL: CU + ALU + registers

New Learning

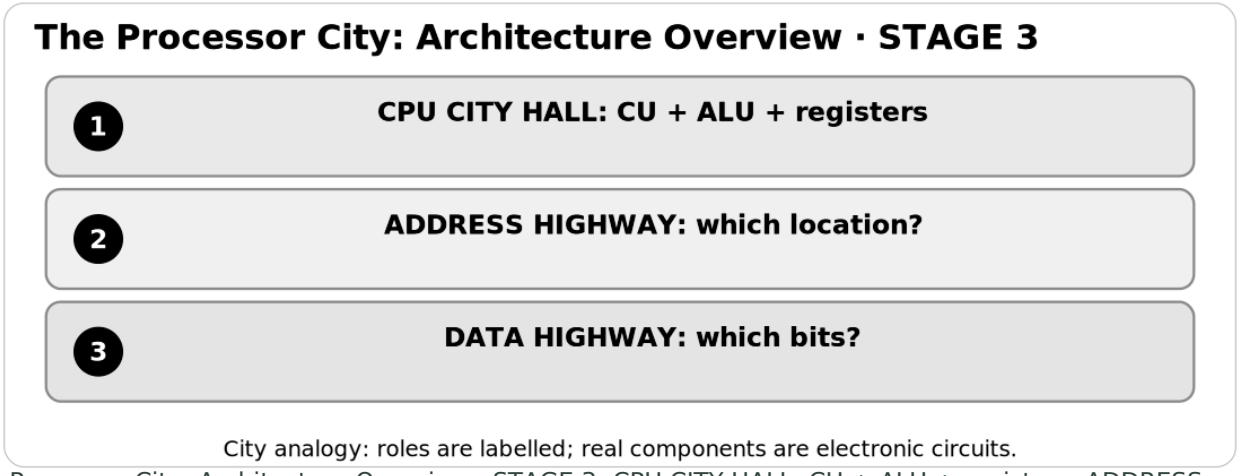


The Processor City: Architecture Overview · STAGE 2; CPU CITY HALL: CU + ALU + registers; ADDRESS HIGHWAY: which location?

New Learning · original lesson diagrams

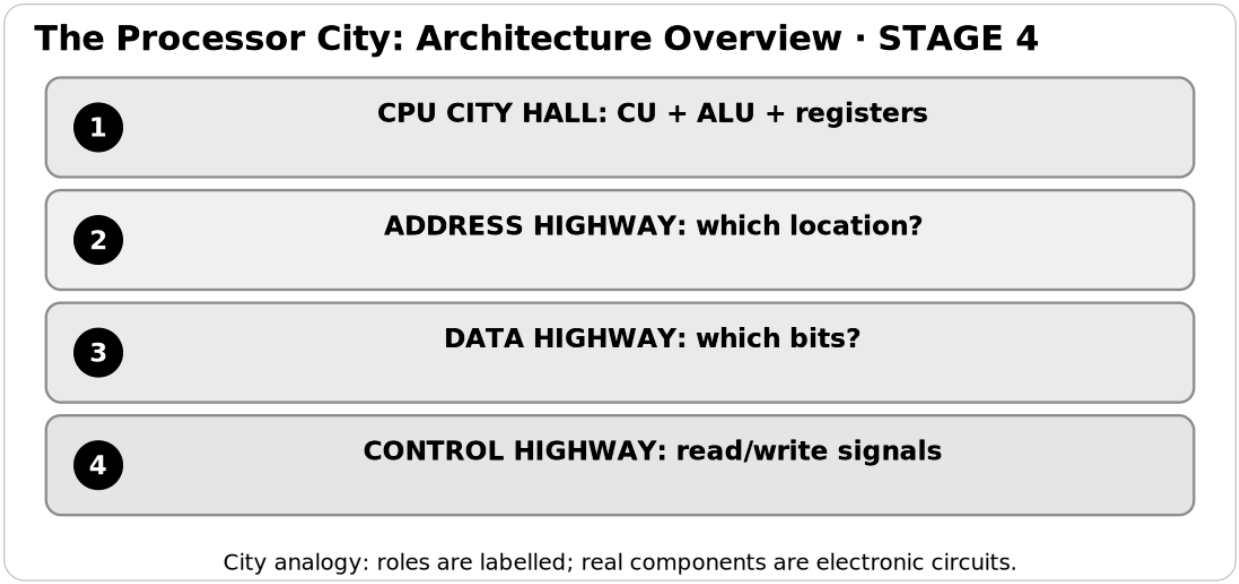
Diagrams appear after the related explanation. Annotate the original source visual.

New Learning



The Processor City: Architecture Overview · STAGE 3; CPU CITY HALL: CU + ALU + registers; ADDRESS HIGHWAY: which location?; DATA HIGHWAY: which bits?

New Learning

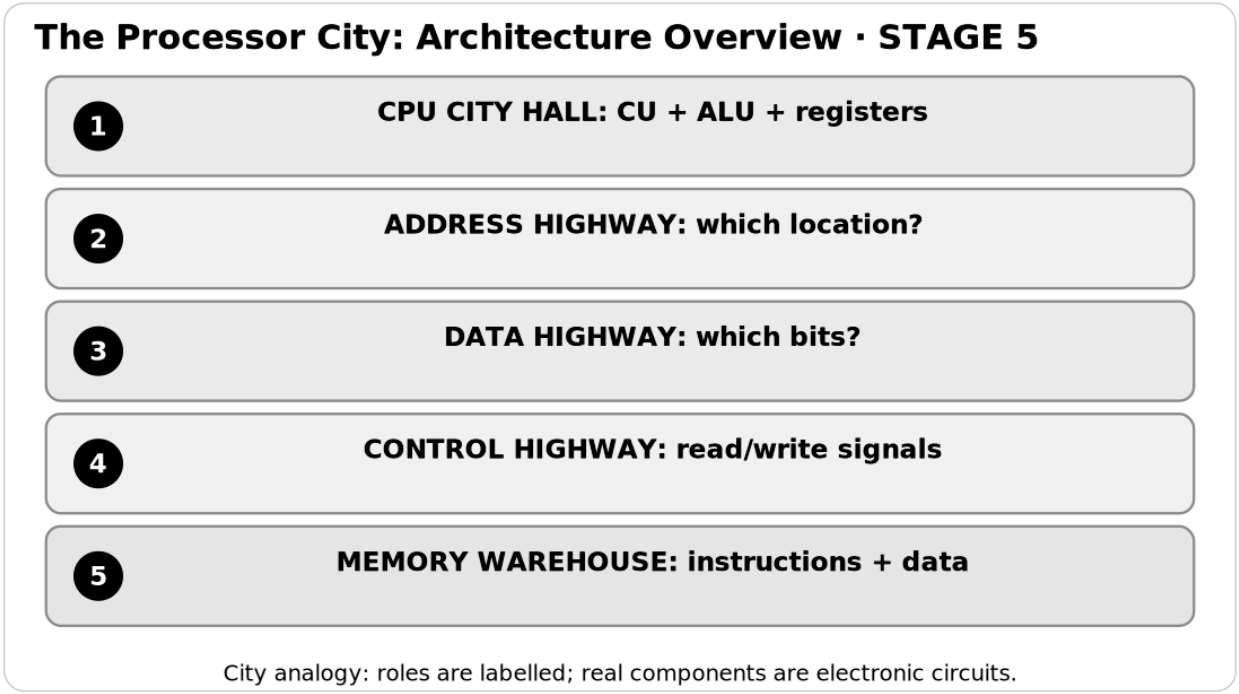


The Processor City: Architecture Overview · STAGE 4; CPU CITY HALL: CU + ALU + registers; ADDRESS HIGHWAY: which location?; DATA HIGHWAY: which bits?; CONTROL HIGHWAY: read/write signals

New Learning · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

New Learning



The Processor City: Architecture Overview · STAGE 5; CPU CITY HALL: CU + ALU + registers; ADDRESS HIGHWAY: which location?; DATA HIGHWAY: which bits?; CONTROL HIGHWAY: read/write signals; MEMORY WAREHOUSE: instructions + data

I DO — WORKED EXAMPLE

Follow the worked method

Explain a memory read using the city analogy and exact bus names.

Model response

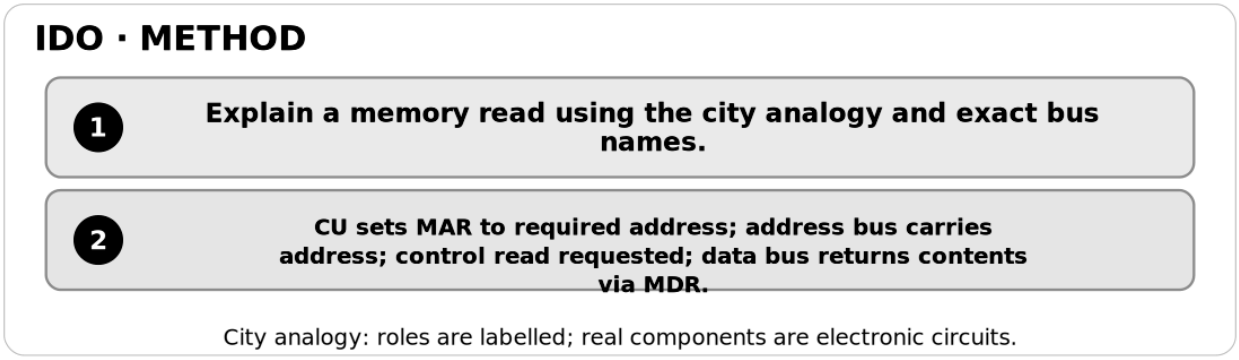
The address identifies where; data is what is transferred. CU coordinates the process.

I Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

Explain a memory read using the city analogy and exact bus names.

The address identifies where; data is what is transferred. CU coordinates the process.



IDO · METHOD; Explain a memory read using the city analogy and exact bus names.; CU sets MAR to required address; address bus carries address; control read requested; data bus returns contents via MDR.

WE DO — GUIDED PRACTICE

Try → Discuss → Improve

TRY: Attempt the guided questions in BLACK PEN. DISCUSS: Compare approaches with a partner, explain your reasoning and agree one improvement. IMPROVE: After the Exit Ticket, use the source CHECK model in GREEN PEN to correct and explain errors.

Guided 1 — TRY

Explain why ALU and CU must be shown as separate parts of City Hall.

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations or a justified explanation.

Question 1

Name and describe ALU, Control Unit, and registers in processor. [6]

Question 2

Using correct names, explain address, data and control bus during memory read. [6]

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations or a justified explanation.

Question 3

Evaluate usefulness and limitation of Processor City analogy for Von Neumann architecture. [6]

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking.

The data bus carries the address of a memory location.

Circle: Confident / Getting there / Need help

Explain what you can now do independently.

GREEN PEN — Correct and explain

Choose one error. Write the correct method and explain precisely why your first response was wrong.

WE DO — CHECK AND IMPROVE

Source-grounded model answers

CHECK only after completing the guided practice and Exit Ticket. Use GREEN PEN to improve your method and explain any error.

Guided 1 — CHECK

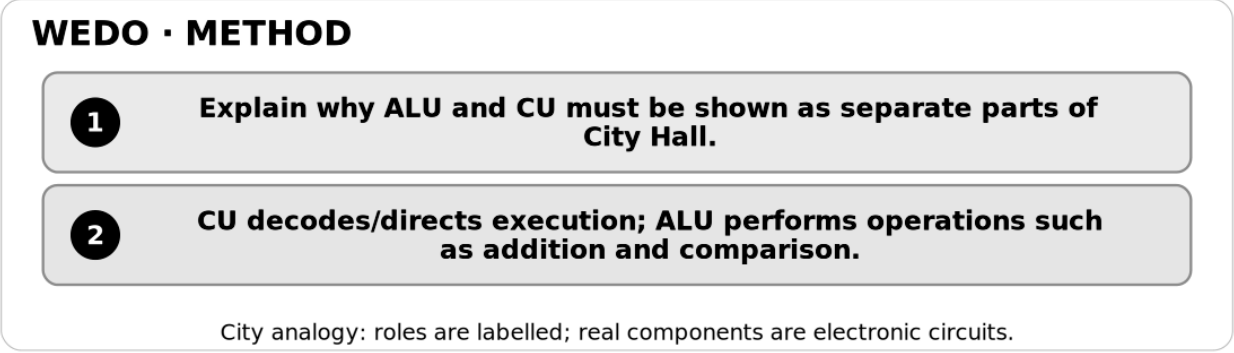
Model response
Saying the CU performs arithmetic confuses separate roles.

We Do source CHECK · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

Explain why ALU and CU must be shown as separate parts of City Hall.

Saying the CU performs arithmetic confuses separate roles.



WEDO · METHOD; Explain why ALU and CU must be shown as separate parts of City Hall.; CU decodes/directs execution; ALU performs operations such as addition and comparison.